

Dimensionless Gauge Couplings as Moduli: Structural Non-Derivability, Renormalization Group Geometry, and Determinability Limits

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Abstract

We establish structural limits on the determinability of dimensionless gauge couplings in renormalizable four-dimensional quantum field theories. We prove that symmetry, locality, unitarity, renormalizability, and anomaly cancellation do not internally fix such couplings. We formalize coupling space as a moduli space of structurally admissible theories, interpret renormalization group flow as motion on theory space, derive perturbative stability basin constraints, classify discrete versus continuous structural data, and show that any purported purely algebraic derivation of a unique dimensionless coupling necessarily introduces additional dynamical, boundary, or empirical structure. We further prove that Gaussian fixed points cannot uniquely determine gauge couplings and state necessary and sufficient fixed-point conditions for genuine unique fixing.

1 Scope and Structural Framework

All results apply at the level of renormalizable four-dimensional quantum field theory absent additional ultraviolet completion, dynamical selection principles, or empirical matching conditions.

We use the term *moduli* in the structural sense of continuous parameters labeling inequivalent but structurally admissible theories. No claim is made about smooth manifold structure.

In intuitive terms, structural consistency determines the form of interactions but not their numerical strength. Renormalization group fixed points reduce freedom only if they eliminate all continuous deformation directions. Absent such strongly isolated fixed points, dimensionless couplings remain continuous parameters specifying location in theory space.

We assume:

1. Lorentz invariance,
2. Local gauge invariance with compact Lie group G ,
3. Power-counting renormalizability in four dimensions,
4. Unitarity,
5. Gauge anomaly cancellation,
6. Canonical kinetic normalization.

2 Structural Non-Derivability

Definition 1. A parameter is internally fixed if structural constraints permit at most one physically inequivalent value.

Theorem 1. No dimensionless gauge coupling g is internally fixed by the structural axioms above.

Proof. Gauge invariance fixes interaction structure but not coefficient magnitude. Renormalizability restricts operator dimension but not coupling strength. Anomaly cancellation constrains representations but does not constrain g . Field redefinitions preserving canonical normalization cannot remove g without altering physical observables. Therefore a continuous family of inequivalent theories exists. $\square$

3 Theory Space and Renormalization Group Geometry

Let θ^a denote renormalizable couplings. The set of physically inequivalent theories forms theory space $\mathcal{T}$. Dimensionless couplings serve as moduli that label inequivalent but structurally admissible theories.

Renormalization group flow is defined by

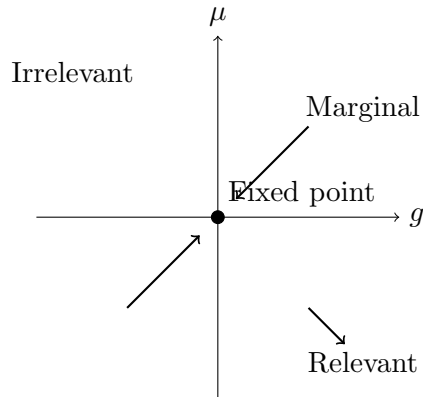
$$\mu \frac{d\theta^a}{d\mu} = \beta^a(\theta).$$

Linearizing near a fixed point θ_* gives

$$\mu \frac{d}{d\mu}(\theta - \theta_*)^a = M^a_b(\theta - \theta_*)^b,$$

where M is the stability matrix. Eigenvalues classify deformations as relevant, irrelevant, or marginal.

Schematic Renormalization Group Structure



4 Structural Classification

Theorem 2. Within the structural framework:

1. Discrete data fixed by structure include the gauge group, global form, representation content, anomaly constraints, and topological sectors.

2. Continuous moduli include gauge couplings, Yukawa couplings, and scalar self-couplings compatible with the structural axioms.

5 Worked Examples

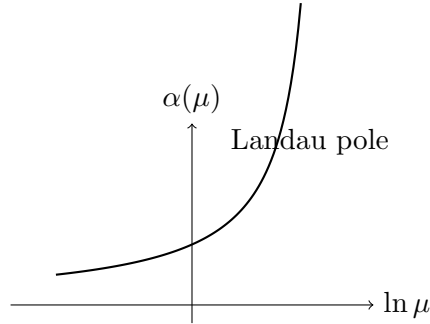
Quantum Electrodynamics

The one-loop running of the fine structure constant is

$$\frac{d\alpha}{d\ln\mu} = \frac{2}{3\pi} \left(\sum_f Q_f^2 \right) \alpha^2.$$

The beta-function coefficient depends only on discrete charge assignments, while the initial value of α remains a modulus.

Landau Pole Illustration



Quantum Chromodynamics and Dimensional Transmutation

The one-loop running of the strong coupling is

$$\frac{dg_s}{d\ln\mu} = -\beta_0 g_s^3, \quad \beta_0 = \frac{1}{16\pi^2} (11N_c - 2N_f).$$

Integrating yields

$$\frac{1}{g_s^2(\mu)} = 2\beta_0 \ln \left(\frac{\mu}{\Lambda_{\text{QCD}}} \right),$$

where

$$\Lambda_{\text{QCD}} = \mu \exp \left(-\frac{1}{2\beta_0 g_s^2(\mu)} \right).$$

Confinement and dimensional transmutation reparameterize continuous freedom but do not eliminate it.

6 Perturbative Stability Basin

For abelian gauge theory,

$$\frac{d\alpha}{d\ln\mu} = b\alpha^2.$$

Requiring validity up to scale Λ gives

$$\alpha(\mu_0) \leq \frac{1}{b\ln(\Lambda/\mu_0)}.$$

Stability restricts viable regions but does not uniquely determine couplings.

7 Fixed Point Determinacy and Uniqueness

Definition 2. A fixed point is strongly isolated if it possesses no relevant and no marginal directions.

Theorem 3. Unique determination of a dimensionless coupling requires a strongly isolated ultraviolet fixed point.

Gaussian fixed points in four dimensions contain at least one marginal direction corresponding to the gauge coupling and therefore cannot uniquely fix it.

8 Domain of Applicability

The analysis applies to renormalizable four-dimensional quantum field theories. Extensions to infinite-dimensional Wilsonian effective field theories or to theories including gravity require additional structural analysis. In enlarged coupling spaces, unique fixing would occur only if a strongly isolated fixed point exists.

9 Conclusion

Structural principles determine admissible theories but not numerical values of dimensionless couplings. Unique fixing requires a strongly isolated fixed point or additional dynamical, boundary, or selection structure. In their absence, dimensionless couplings are moduli specifying location in theory space.